

Experiment 2: Crypt-Arithmetic Problem

Aim

To solve the Crypt-Arithmetic problem using Python.

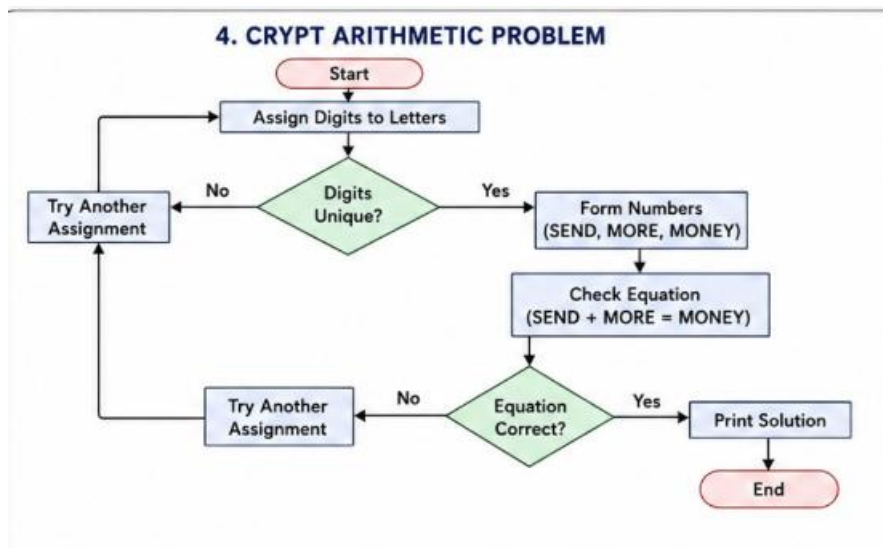
Objective

To assign digits to letters so that the arithmetic equation is satisfied.

Algorithm

1. Take the words in the equation.
2. Assign unique digits to each letter.
3. Check if the equation is valid.
4. Display the solution.

Flowchart



CODE

```
from itertools import permutations
```

```
letters = ('S','E','N','D','M','O','R','Y')
```

```
for p in permutations(range(10), 8):
```

```
    d = dict(zip(letters, p))
```

```
if d['S']==0 or d['M']==0:
```

```
    continue
```

```
send = d['S']*1000+d['E']*100+d['N']*10+d['D']
```

```
more = d['M']*1000+d['O']*100+d['R']*10+d['E']
```

```
money = d['M']*10000+d['O']*1000+d['N']*100+d['E']*10+d['Y']
```

```
if send + more == money:
```

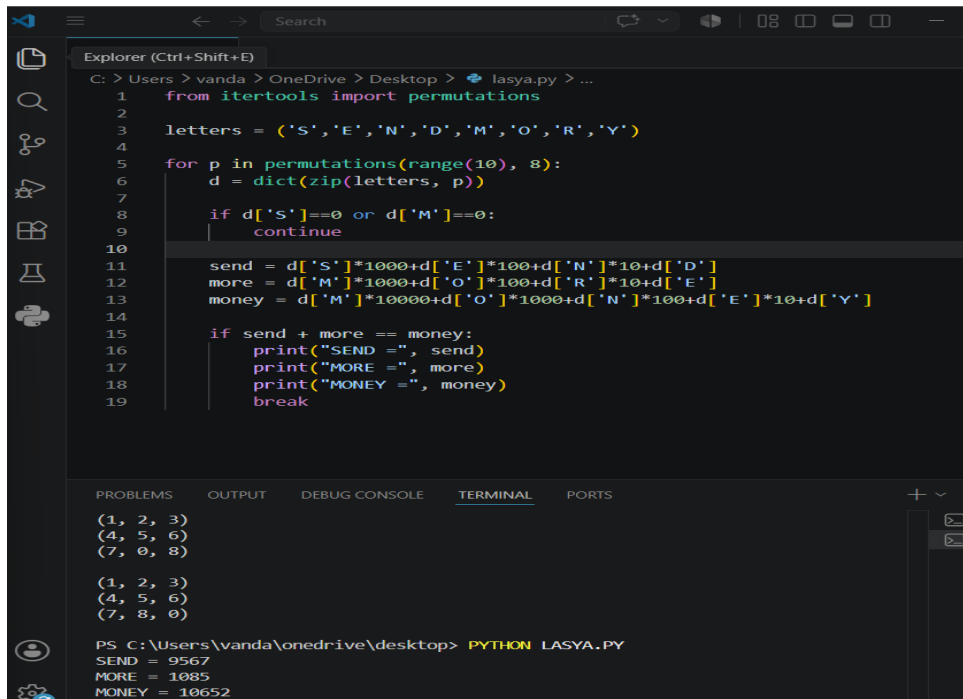
```
    print("SEND =", send)
```

```
    print("MORE =", more)
```

```
    print("MONEY =", money)
```

```
    break
```

Output:



```
1 from itertools import permutations
2
3 letters = ('S','E','N','D','M','O','R','Y')
4
5 for p in permutations(range(10), 8):
6     d = dict(zip(letters, p))
7
8     if d['S']==0 or d['M']==0:
9         continue
10
11     send = d['S']*1000+d['E']*100+d['N']*10+d['D']
12     more = d['M']*1000+d['O']*100+d['R']*10+d['E']
13     money = d['M']*10000+d['O']*1000+d['N']*100+d['E']*10+d['Y']
14
15     if send + more == money:
16         print("SEND =", send)
17         print("MORE =", more)
18         print("MONEY =", money)
19         break
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

PS C:\Users\vanda\onedrive\desktop> PYTHON LASYA.PY
SEND = 9567
MORE = 1085
MONEY = 10652
```

Result

The Crypt-Arithmetic problem was solved successfully.

Conclusion: The program assigns digits to letters and finds the correct arithmetic solution.

